

Personal AI Research OS — Architecture v3

This document defines the revised architecture for a personal AI-native investment and research operating system. The system is optimized for durable knowledge accumulation, investment research, coding, data analysis, and structured cognitive workflows.

1. Core Vision

The system is designed as a high-signal cognitive operating environment. The objective is not to maximize AI output volume, but to compound intelligence over time through: structured memory, reproducible workflows, adversarial reasoning, and disciplined research practices.

The central architectural principle is: Claude should function as the primary operational cockpit. The user should feel like they are operating from one coherent environment rather than orchestrating fragmented tools.

2. Core Architectural Principles

- Claude is the primary cockpit and operating environment.
- GitHub acts as canonical machine memory.
- Notion acts as human-readable reflective memory.
- ChatGPT acts as a cognitive sparring partner and external reviewer.
- Agents are operational cognitive modes, not autonomous AI workers.
- High-stakes outputs require mandatory adversarial review.
- Memory must be selective and anti-rot by design.
- The system prioritizes reproducibility and traceability.
- Complexity must emerge from validated need, not speculation.

3. Consolidated Operating Model

Component	Role	How It Should Feel
Claude	Primary operating cockpit	Main workspace
GitHub	Canonical machine memory	Structured state
Notion	Human reflective memory	Readable notebook
Cursor	Implementation environment	Engineering workstation
Terminal/APIs	Execution and data layer	Reality interface
ChatGPT	Independent sparring partner	External challenger

Operationally, the user should spend most working time inside Claude Co-Work. GitHub, Cursor, Notion, APIs, and terminal workflows should support the Claude-centered workflow rather than compete with it.

4. Cognitive Layer

Claude and ChatGPT should not be treated as interchangeable systems. Claude is the primary operating environment. ChatGPT acts as a deliberate independent sparring and review layer.

Use Claude For	Use ChatGPT For
Long-running project continuity	Independent second opinions
Architecture and workflow design	Alternative framings
Structured research synthesis	Adversarial critique
Memory-aware reasoning	Detecting blind spots
Operational coordination	Prompt refinement

5. Agent System

Agents are not separate autonomous systems. They are operational cognitive modes invoked intentionally inside Claude. One consolidated Co-Work environment should be used initially.

Agent	Role	Primary Function
Atlas	Research Lead	Research synthesis, framing, investigation coordination
Vector	Quant & Data	Modeling, data analysis, statistical reasoning
Forge	Systems & Implementation	APIs, SDKs, infra feasibility, implementation architecture
Beacon	Investment Strategist	Investment thesis and portfolio reasoning
Sentinel	Red Team / Critic	Adversarial review and assumption attacks
Archive	Documentation Steward	Memory hygiene and continuity
Compass	Legal / Structure	Legal and structural analysis

Default agent: Atlas (Research Lead).

Forge is a critical addition. The system is not only an investment research environment but also a software and infrastructure effort. Forge is responsible for: API understanding, SDK feasibility, systems integration, tooling constraints, engineering tradeoffs, implementation architecture, debugging strategy, and platform reasoning.

6. Agent Invocation Model

- Agents are invoked explicitly inside the same Claude Co-Work environment.
- Agent definitions live in GitHub under docs/agents/.
- Claude Projects preload foundational operating documents.
- One conversation may involve multiple agents.
- Agent transitions should be explicit.
- Sentinel review is mandatory for high-stakes outputs.
- Archive runs at meaningful session close.

7. Memory Architecture

Memory Layer	Lives Where	Purpose
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Machine Memory	GitHub	Canonical structured state
Human Memory	Notion	Readable reflection and navigation
Working Context	Claude Projects / active session	Current reasoning context
Ephemeral Exploration	Temporary only	Discarded unless promoted

Only durable insights should be promoted into GitHub: architecture decisions, validated assumptions, workflows, research conclusions, governance changes, and reusable knowledge.

Notion should not become canonical operational state. It should function as: - reflective layer, - dashboard, - readable summaries, - navigation surface, - and human-oriented review space.

8. Session Lifecycle

```
flowchart TD
    User --> Claude
    Claude --> Atlas
    Atlas --> Vector
    Atlas --> Forge
    Atlas --> Beacon
    Beacon --> Sentinel
    Sentinel --> HumanDecision
    HumanDecision --> Archive
    Archive --> GitHub
    Archive --> Notion
```

Every meaningful session should end with: - session summary, - decisions made, - key learnings, - open questions, - proposed memory updates, - discard decisions, - and next actions.

9. Repository Structure

```
research-os/
  docs/
    architecture/
    governance/
    agents/
  research/
    active/
    completed/
  decisions/
    adr/
    investment/
  logs/
    sessions/
  templates/
  code/
  archive/
```

10. Integration Philosophy

Integration	Recommendation	Timing
GitHub	Essential	Day 1
Cursor	Essential	Day 1
Terminal	Essential	Day 1
APIs	Essential	Day 1
Notion	Recommended	Early
Drive	Reference material	Early

Sheets	Quick modeling	Early
Slack	Avoid initially	Later or never
Advanced orchestration	Avoid	Only if clearly necessary

Slack is intentionally deprioritized. Using Slack as an “agent collaboration layer” is likely premature and risks: fragmentation, theater, operational noise, and duplicated context.

11. Governance

- Human remains final authority.
- High-stakes outputs require Sentinel review.
- GitHub commits should remain human-reviewed initially.
- Architecture changes require ADRs.
- Memory updates must avoid rot and duplication.
- No autonomous execution without explicit approval.

12. Recommended Immediate Execution

The correct next step is not advanced automation. The correct next step is establishing a stable operational rhythm.

- Create dedicated macOS research user
- Create private GitHub repo
- Create one consolidated Claude Co-Work project
- Upload architecture documents
- Create foundational markdown structure
- Run first 5–10 real research sessions
- Iterate from observed friction

13. Final Recommendation

The biggest risk now is premature complexity. The system should evolve through real usage patterns rather than speculative infrastructure design.

The architecture should optimize for: clarity, continuity, cumulative intelligence, reproducibility, and durable deep work.